COMPUTER AIDED ENGINEERING GRAPHICS

A9304

Experiment No. 5

Projection of Lines - I

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I B.Tech I Semester

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Affiliated to **JNTUH**, Approved by **AICTE**, Accredited by **NAAC** with **A++** Grade, **ISO 9001:2015** Certified

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Projection of Lines I

1 Objectives

To perform orthographic projections for Lines situated in different orientations.

2 Introduction

A straight line is defined as the locus of a point which moves unidirectionally. The straight line can also be defined as the shortest distance between two points. The projections of straight lines can be drawn by joining the respective projections of its end points. The word 'line' is used in text for straight lines for the sake of simplicity. The actual length of the line is commonly called true length and is denoted by T.L.

2.1 Orientation of straight Line

A straight line may be in one of the following positions.

1. Line parallel to both horizontal plane (H.P.) and vertical plane (V.P.).
2. Line perpendicular to H.P. (and parallel to V.P.).
3. Line perpendicular to V.P. (and parallel to H.P.).
4. Line inclined to H.P. and parallel to V.P.
5. Line inclined to V.P. and parallel to H.P.
6. Line situated on H.P.
7. Line situated in V.P.
8. Line situated on both H.P. and V.P. (i.e., on the reference line, xy).
9. Line inclined to both the reference planes.
    - Line inclined to both H.P. and V.P. such that $\theta + \phi < 90^0$.
    - Line inclined to both H.P. and V.P. such that $\theta + \phi = 90^0$.

Projections of a straight line lying in the first angle shall have its front view above xy and the top view below xy.

3 Objectives

1. Understand the fundamental principles of orthographic projection of straight lines with respect to the horizontal plane (H.P.) and vertical plane (V.P.), including front view, top view, reference line, projectors, and true length.
2. Apply AutoCAD drawing and construction commands to draw the projections of lines that are parallel to both H.P. and V.P., perpendicular to H.P., and perpendicular to V.P.
3. Analyze the front and top views of a line to identify its spatial orientation and evaluate whether the drawn projections satisfy the given geometrical conditions.

4 Pre-Lab Questions

Q1. Explain the Concept of Layers in AUTOCAD.

Q2. what is a Projector?

Q3. Explain the Process of Plotting?

5 Aim

Perform the following in AUTOCAD

- A 50mm long line PQ is parallel to both the H.P. and the V.P. It is 25mm in front of the V.P. and 60mm above the H.P. Draw its projections.
- A 60mm long line AB has its end A 20mm above H.P. The line is perpendicular to the H.P. and 40mm in front of the V.P. Draw its projections
- A 60mm long line CD has its end C 20mm in front of the V.P. The line is perpendicular to the V.P. and 40mm above the H.P. Draw the projections of the line

6 Software Used

- AUTOCAD

7 System Requirements

1. **System Type:** Windows 10 64 Bit Operating System
2. **Processor:** i3
3. **RAM:** 4 GB

8 Commands Used

Line, Text, Layers, Dimension Linear.

9 Experimental procedure

10 Learning Outcomes

At the end of this experiment, the student will be able to:

LO1. Recall the terms H.P., V.P., XY reference line, front view, top view, projector, and true length.

LO2. Explain how the orientation of a line relative to H.P. and V.P. affects the position and appearance of its front and top views.

LO3. Construct accurate AutoCAD projections for the given line problems using appropriate drawing commands and drafting standards.

LO4. Differentiate between lines parallel to both planes, perpendicular to H.P., and perpendicular to V.P. by analyzing their orthographic views.

LO5. Verify the correctness of the drawn projections by checking view alignment, distances from the XY line, and point-view/true-length conditions.

11 Post-Lab Questions

Q1. In a AutoCAD drawing, both the front view and the top view of a line lie exactly on the XY reference line. What is the position of this line in space? How many views are needed to represent it?

Q2. In a AutoCAD drawing, the front view of a line is a line parallel to XY and the top view is also a line parallel to XY. Both views are of equal length. What is the orientation of this line in space?

Q3. In the front view of a AutoCAD drawing, a line appears as a point. In the top view, it appears as a vertical line. Identify the orientation of this line.

Q4. In the top view of a 2D AutoCAD drawing, a line appears as a point. In the front view, it appears as a vertical line. Identify the orientation of this line.

12 Result

The solutions of the projection of lines are analysed and drawn in the AUTOCAD.

13 Solutions

Note: These solutions are provided for reference purposes only and are not intended to be copied into the record.

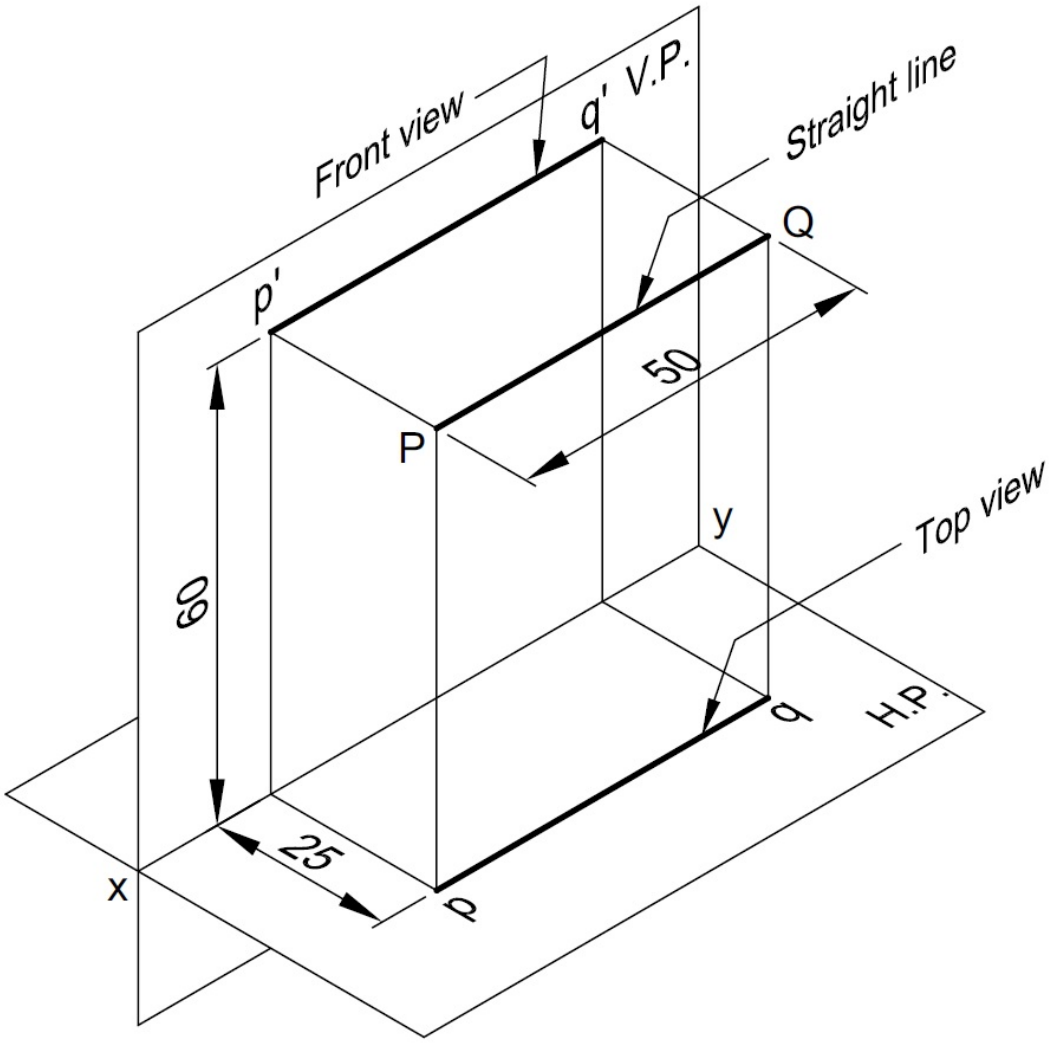


Fig. 1: a. explanation b. Solution

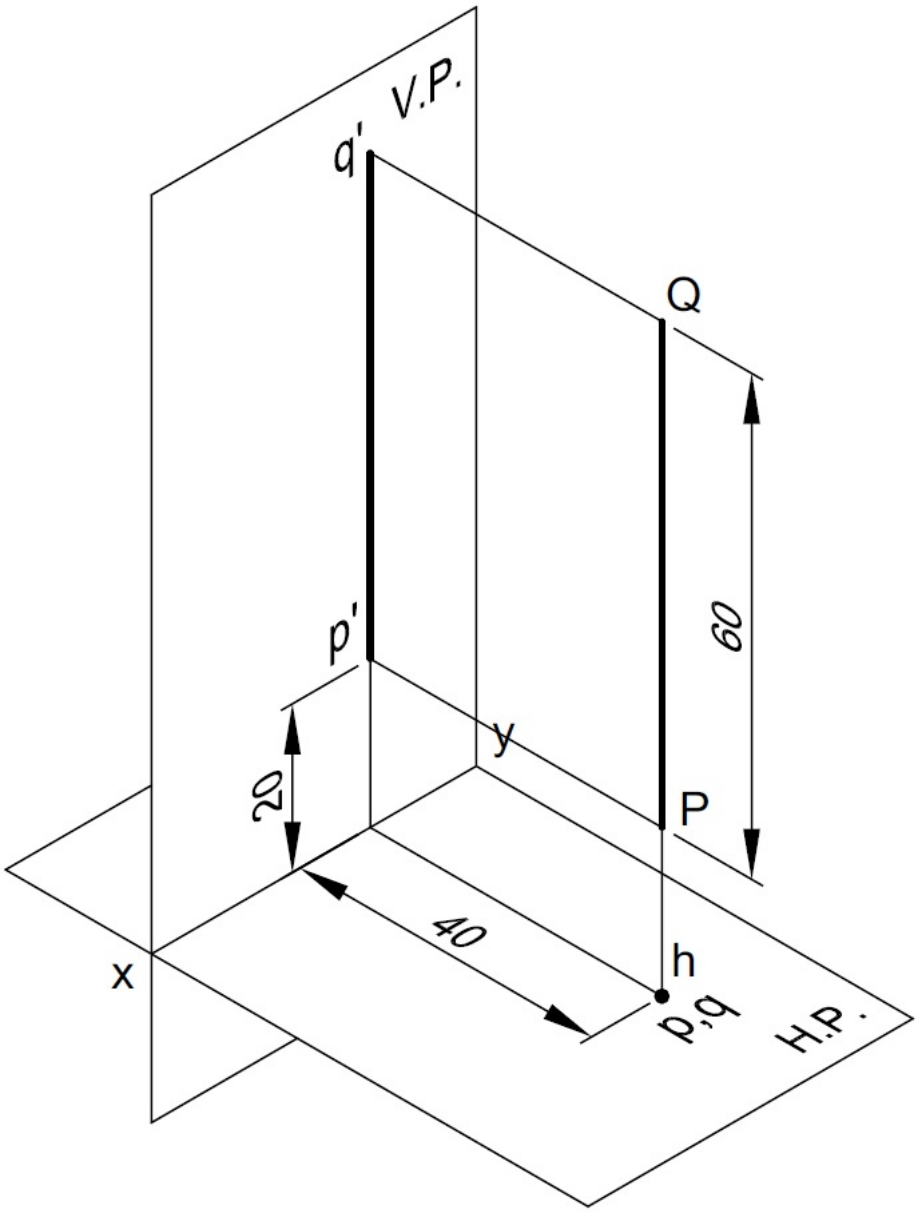


Fig. 2: a. explanation b. Solution

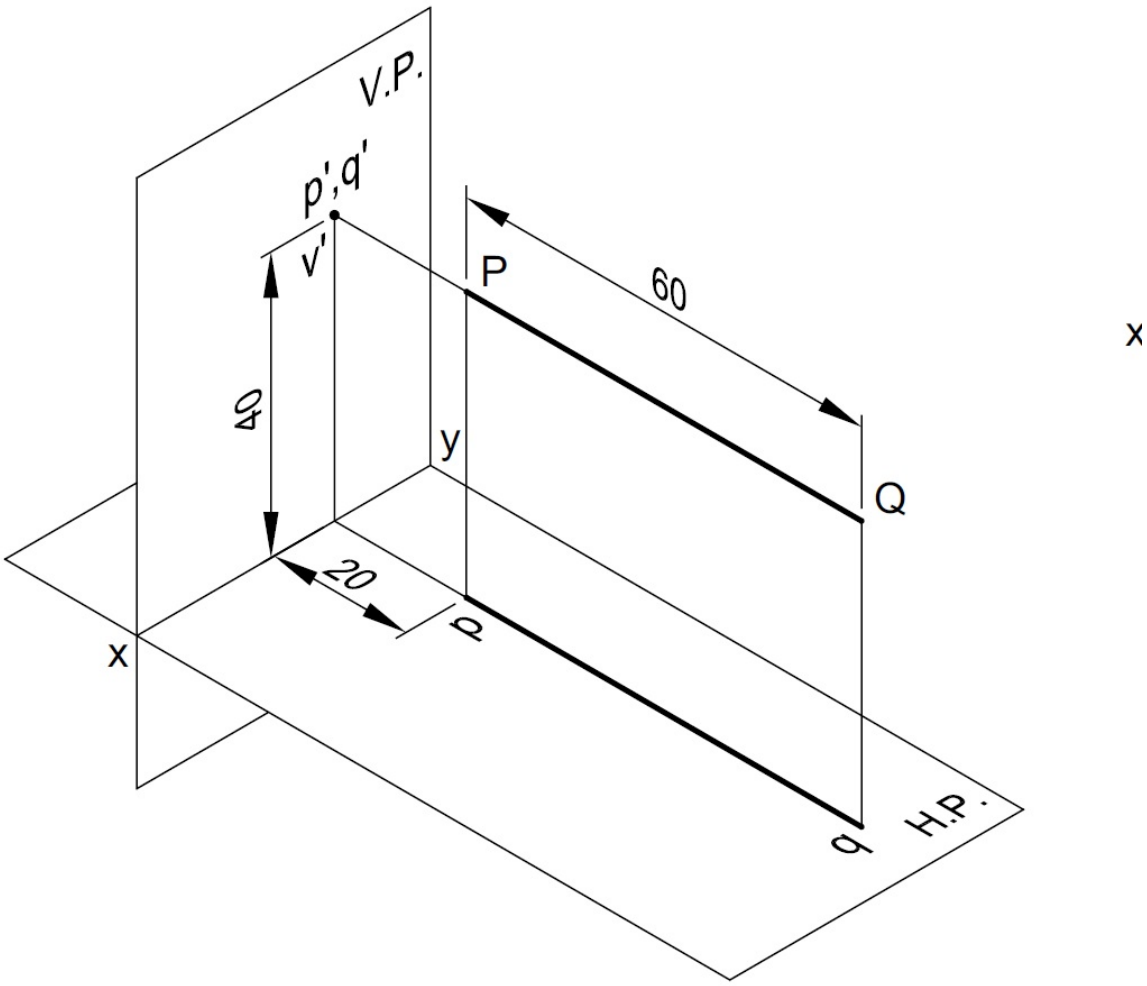


Fig. 3: a. explanation b. Solution